Hints: Number theory – 1

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Fix m. Then $\sigma(k) = mk$ is a permutation of $\mathbb{Z}/n$, and you need a permutation π with $\pi^2 = \sigma$. When does a permutation have a square root? Look at its cycle type.

Hint 2. A permutation is a square exactly when, for every even ℓ, it has an even number of ℓ-cycles. For $n \equiv 0 \pmod 4$ take $m = -1$: it has two fixed points, 0 and $n/2$, and $(n-2)/2$ transpositions, an odd number. For odd $n > 1$ pick $p^e \parallel n$ and m (by CRT) a primitive root mod p^e with $m \equiv 1 \pmod{n/p^e}$. The cycles of length $\varphi(p^e)$ are exactly the orbits of the k that are units mod p^e, and there are n/p^e of them, an odd number.

Hint 3. For $n = 2n'$ with n' odd, $\mathbb{Z}/n \cong \mathbb{Z}/2 \times \mathbb{Z}/n'$ and m fixes the first coordinate. So σ is two copies of the same permutation, and every cycle length occurs an even number of times. Answer: $n = 1$ and all $n \equiv 2 \pmod 4$.

Problem 2

Hint 1. Primes work: the sum is $x + x^p - 2x$. Now let n be composite and square-free, and work modulo its largest prime factor p (it is odd). Write $n = pm$.

Hint 2. The divisors of n are d and pd with $d \mid m$, and $x^{pd} \equiv x^d \pmod p$. So the condition reads $p \mid 2(\sum_{d \mid m} x^d - \tau(m)x)$ for all x.

Hint 3. Reduce each exponent mod $p-1$ into $[1, p-1]$. You get a polynomial of degree $< p$ that vanishes on $\mathbb{F}_p$, so it is zero. Comparing the coefficients of x forces every $d \mid m$ to satisfy $d \equiv 1 \pmod{p-1}$. A prime $q \mid m$ is $< p$, a contradiction. Answer: the primes (and trivially $n = 1$).

Problem 3

Hint 1. Yes. Build $a_1 = 1 < a_2 < \dots$ inductively. Look for $a_{n+1} = x$ with $x \equiv 1 \pmod{M^2}$, where $M = \prod_{q \leqslant n} q$ over primes q.

Hint 2. For a prime $q \leqslant n$ we get $x + a_i \equiv 1 + a_i = a_1 + a_i \pmod{q^2}$, which is square-free by induction, so $q^2 \nmid x + a_i$. Only primes $q > n$ remain.

Hint 3. Count $x = 1 + kM^2$, $1 \leqslant k \leqslant K$. For $q > n$ a fixed a_i excludes at most $K/q^2 + 1$ values of k, and only $q \leqslant \sqrt{2KM^2}$ matter. The bad k number at most $nK \sum_{q>n} q^{-2} + O(nM\sqrt{K})$. Since $\sum_{q>n,\ q \text{ odd}} q^{-2} < 1/(2(n-1))$, this is $< K$ for large K.

Problem 4

Hint 1. A sequence is determined by (a_1, a_2) via $a_{n+2} = (1 + a_{n+1})/a_n$. Compute a few terms symbolically.

Hint 2. The recurrence is periodic with period 5: $a_3 = \frac{1+b}{a}$, $a_4 = \frac{1+a+b}{ab}$, $a_5 = \frac{1+a}{b}$, $a_6 = a$. So the shift $(a_1, a_2) \mapsto (a_2, a_3)$ has order dividing 5 on the set of admissible pairs.

Hint 3. Orbits have size 1 or 5, so $f(p)$ is congruent mod 5 to the number of fixed points: $a_1 = a_2 = a$ with $a^2 = a + 1$. The discriminant is $5 \neq 0 \bmod p$, so there are 0 or 2 such a.

Problem 5

Hint 1. Expand $f(a+tp^k) = f(a) + tp^k f'(a) + p^{2k}(\dots)$. Since $2k \geqslant k+1$, the condition is the linear congruence $f(a)/p^k + tf'(a) \equiv 0 \pmod{p}$, which has a unique solution t.

Hint 2. $x^2 \equiv 2 \pmod 7$ has $x \equiv 3, 4$, and $f'(3) = 6 \not\equiv 0$. Lift twice. Answer: $x \equiv 108, 235 \pmod{343}$. For $p \equiv 1 \pmod 4$, a root of $x^2 \equiv -1 \pmod p$ exists, e.g. $((\frac{p-1}{2})!)^2 \equiv -1$ by Wilson, and $f'(a) = 2a \not\equiv 0$.

Hint 3. Failure: $f(x) = x^2 - 3$, $p = 2$. Here $f(1) \equiv 0 \pmod 2$ but $f'(1) \equiv 0$, and $x^2 \equiv 3 \pmod 4$ has no solution.

Problem 6

Hint 1. Choose distinct primes $p_1, \dots, p_n$ and solve $x + i \equiv 0 \pmod{p_i^2}$ for $i = 1, \dots, n$ by CRT.

Hint 2. For the second claim use $2n$ distinct primes and solve $x + i \equiv 0 \pmod{p_i q_i}$. Each $x + i$ then has two different prime factors, so it is not a prime power.

Problem 7

Hint 1. Pair k with $p - k$: $\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)}$. This already gives divisibility by p.

Hint 2. So $H_{p-1} = \frac{p}{2} \sum_{k=1}^{p-1} \frac{1}{k(p-k)}$, and modulo p we have $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2}$. It remains to show $p \mid \sum k^{-2}$ (as a p-integral rational).

Hint 3. $k \mapsto k^{-1}$ permutes the nonzero residues, so $\sum k^{-2} \equiv \sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod p$ for $p \geqslant 5$.

Problem 8

Hint 1. If f works, so do $f + c$ and $-f$. Normalise $f(0) = 0$. Then $f(p) \mid p^n$ for every prime p, so $f(p) = \pm p^{d_p}$.

Hint 2. By pigeonhole there are ε, d with $f(p) = \varepsilon p^d$ for infinitely many primes p. Normalise $\varepsilon = 1$; $d = 0$ is impossible. Write $n = md + r$ with $0 \leqslant r < d$.

Hint 3. For such p and any x: $p^d - f(x) \mid p^n - x^n$, and $p^n = p^r (p^d)^m \equiv p^r f(x)^m$. So $p^d - f(x) \mid p^r f(x)^m - x^n$. For large p this forces $p^r f(x)^m = x^n$, hence $r = 0$ and $f(x) = x^d$ (m is odd). Answer: $f(x) = \pm x^d + c$ with $d \mid n$, $d \geqslant 1$.

Problem 9

Hint 1. Substitute $n = 0$ and $m = 0$, and then $(m, n) \to (n, m)$. Show that every $f(k)$ divides $f(0)$ and that $f(-k) = f(k)$.

Hint 2. Put $x = f(m) \leqslant y = f(n)$ and $z = f(n - m)$. The pairs (n, m), $(n, n - m)$, $(m, m - n)$ give $z \mid y - x$, $x \mid y - z$ and $y \mid x - z$.

Hint 3. If $z > y$, then $0 \leqslant y - x < z$ forces $x = y$. If $z \leqslant y$, then $|x - z| < y$ forces $x = z$, and then $x \mid y - x$. In both cases $x \mid y$.

Problem 10

Hint 1. Suppose $x > 1$. If a prime $q \equiv 3 \pmod 4$ divides some $2^N y + 1$, show that $q \mid x^2 - 1$. Then it suffices to find infinitely many such primes.

Hint 2. If $q \mid 2^N y + 1$, then $q \mid 2^{N+t(q-1)} y + 1$ for all t. So $x^{2^M} \equiv 1 \pmod q$ for arbitrarily large M, and $\operatorname{ord}_q(x)$ divides $\gcd(2^M, q - 1) = 2$.

Hint 3. Infinitely many such q: WLOG y is odd (absorb powers of 2 into N). If S is the finite set of all such primes, choose $N \equiv 1$ modulo every $\varphi(q^{e_q+1})$, where $e_q = v_q(2y+1)$. Then $2^N y + 1$ and $2y + 1$ have the same q-parts for $q \in S$ and no other prime factors $\equiv 3 \pmod 4$. Compare both numbers mod 4: $2y + 1 \equiv 3$, but $2^N y + 1 \equiv 1$.